

M5STACK

M5PM1 Chip User Manual

V 1.9

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I. Overview

M5PM1 is a power management IC programmed with M5Stack custom power management firmware, designed to provide fixed implementations of power port control, charging control, timed wake-up, and other power management functions.

1 Resource

- (1) 5 multiplexed GPIOs
- (2) 1 set of I²C interfaces
- (3) 32-bit timer
- (4) 32-byte RAM protected area

2 Function

- (1) 5 GPIOs with extended functions :
 - 2 channels multiplexed as 12-bit ADC
 - 2 channels multiplexed as PWM
 - 1 channel multiplexed for LED control (RGB565)
- (2) GPIO supports programmable pull-up/down resistors, open-drain / push-pull output, and interrupt polarity control
- (3) Supports reading the built-in temperature sensor and internal reference voltage
- (4) I²C interface supports 100 kHz (default) / 400 kHz modes, with address 0x6E
- (5) Supports PWM-based AW8737A control for audio signal amplitude
- (6) Supports driving up to 32 Neopixel RGB LEDs simultaneously

3 Custom firmware pin arrangement

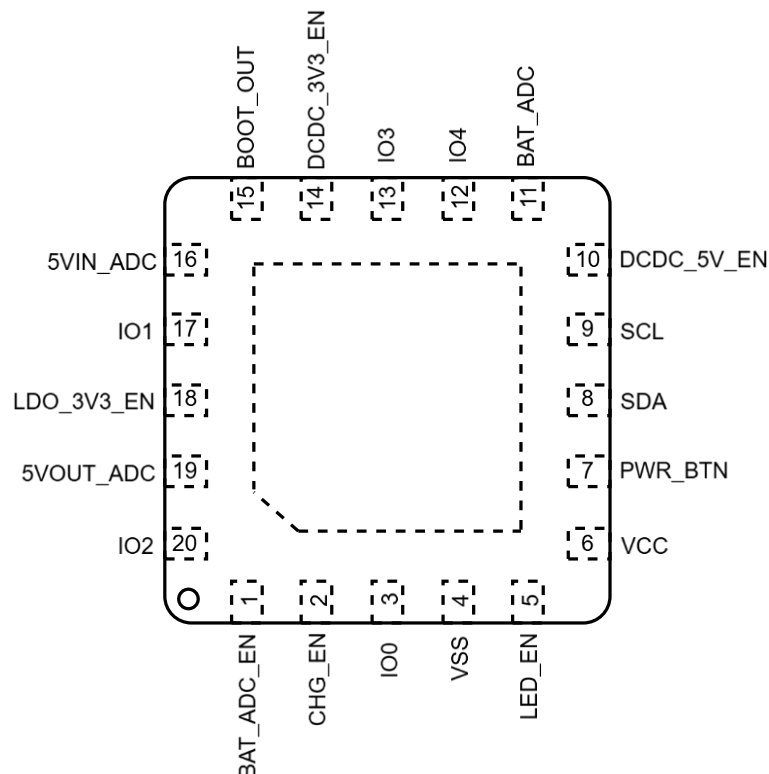


Figure 1 M5PM1 Pin Diagram



Table 1 Custom Firmware Pin

Pin Num	Pin Name	Pin Type	Pull Up/Down
1	BAT_ADC_EN	Push-Pull Output	None
2	CHG_EN	Push-Pull Output	None
3	IO0	GPIO	None
4	VSS	Power	None
5	LED_EN	Push-Pull Output	None
6	VCC	Power	None
7	PWR_BTN	Input	Pull up
8	SDA	I ² C	None
9	SCL	I ² C	None
10	DCDC_5V_EN	Push-Pull Output	None
11	BAT_ADC	ADC	None
12	IO4	GPIO	None
13	IO3	GPIO	None
14	DCDC_3V3_EN	Push-Pull Output	None
15	BOOT_OUT	Open-Drain Output	None
16	5VIN_ADC	ADC	None
17	IO1	GPIO	None
18	LDO_3V3_EN	Push-Pull Output	None
19	5VOUT_ADC	ADC	None
20	IO2	GPIO	None



II. Pin Definitions

Table 2 Pin Definition

Pin	Description	Default	MUX	Note
IO0	GPIO port 0, supports wake-up	GPIO	Neopixel output	Wake-up is mutually exclusive with IO2
IO1	GPIO port 1	GPIO	ADC1	-
IO2	GPIO port 2, supports wake-up	GPIO	ADC2	Wake-up is mutually exclusive with IO0
IO3	GPIO port 3, supports wake-up	GPIO	PWM1	Wake-up is mutually exclusive with IO4
IO4	GPIO port 4, supports wake-up	GPIO	PWM2	Wake-up is mutually exclusive with IO3
BAT_ADC_EN	Battery sampling enable, active high	Sampling enable	-	High level by default
CHG_EN	Battery charging enable, active high	Charging enable	-	High level by default
DCDC_5V_EN	5V DC/DC control, active high	DC control	-	Low level by default
DCDC_3V3_EN	3.3V DC/DC control, active high	DC control	-	High level by default
LDO_3V3_EN	3.3V LDO control, active high	LDO control	-	High level by default
5VIN_ADC	5V input ADC sampling pin	ADC	-	Voltage divider ratio 1:1
5VOUT_ADC	5V output ADC sampling pin	ADC	-	Voltage divider ratio 1:1
BAT_ADC	Battery voltage ADC sampling pin	ADC	-	Voltage divider ratio 1:1
PWR_BTN	Power control button input	Button detection	-	Pull-up enabled by default
LED_EN	Status indicator LED control	LED control		High level by default
BOOT_OUT	Controls main controller ESP32 BOOT	BOOT	-	High level by default
SDA	I ² C data line	I ² C	-	Open-drain mode, external pull-up resistor required
SCL	I ² C clock line	I ² C	-	Open-drain mode, external pull-up resistor required

Note:

1. All GPIO output types default to open-drain mode, including Neopixel driving, PWM output, etc. If no external pull-up resistor is connected, the pin must be configured as push-pull mode to output correctly.
2. LED uses M5PM1 proprietary LED self-control logic and is not recommended for other purposes.



III. Register Mapping

Table 3 Register Map

Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
Device_ID	System	0x00	[7:0]	R	0x50	Device Type	—	—	—
Device_Model	System	0x01	[7:0]	R	0x20	Device Model	—	—	—
HW_REV	System	0x02	[7:0]	R	0x05	Hardware Version	—	—	—
SW_REV	System	0x03	[7:0]	R	0x06	Firmware Version	—	—	—
PWR_SRC	System	0x04	[7:3] Reserved [2:0] VALID	R	—	Power Source Bitmap	—	—	—
WAKE_SRC	System	0x05	[7] Reserved [6:0] FLAGS	R/W	—	Wake-up Source Flag	—	—	—
PWR_CFG	System	0x06	[7:5] Reserved [4] LED CONTROL [3] 5VIN/OUT [2] 3.3V_LDO_EN [1] 3.3V_DCDC_EN [0] CHG_EN	R/W	0x17	Power Management Bit	0b0001011x Charging status is not affected by reset	0b0001011x Charging status is not affected by reset	0b0001011x Charging status is not affected by reset
HOLD_CFG	System	0x07	[7] Reserved [6] 5VIN/OUT [5] 3.3V LDO [4:0] GPIO4~0	R/W	0x00	As corresponding bit is set to 1, the states of corresponding GPIO, LDO, and 5VIN/OUT will be retained after power-off. This register will be reset to 0x00. when Download mode or a Reset is triggered (I ² C watchdog reset, command reset, and user timer reset),	0x00	0x00	—



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Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
BATT_LVP	System	0x08	[7:0]	R/W	0x40	Low-voltage threshold: 2000 mV + n×7.81 mV	—	—	—
I2C_CFG	System	0x09	[7:5] Reserved [4] SPD [3:0] SLP_TO	R/W	0x00	—	—	—	—
WDT_CNT	System	0x0A	[7:0]	R/W	0x00	Watchdog countdown (seconds); set to 0 to disable the watchdog function	—	—	—
WDT_KEY	System	0x0B	[7:0]	W	—	Write 0xA5 to clear and reload	—	—	—
SYS_CMD	System	0x0C	[7:4] KEY(0xA) [3:2] Reserved [1:0] CMD	W	—	System command	—	—	—
—	—	0x0D-0x0F				Reserved	—	—	—
GPIO_MODE	GPIO	0x10	[7:5] Reserved [4:0] GPIO4~0	R/W	0x00	1 = Output, 0 = Input (Must take effect only when the related GPIO_FUNC is set to 00)	0x00	0x00	0b000xxxxx It is determined by bits 0 to 4 of the GPIO_Power_Hold register (0x07): 1 = Hold, 0 = State reset
GPIO_OUT	GPIO	0x11	[7:5] Reserved [4:0] GPIO4~0	R/W	0x00	1=High, 0=Low (Must take effect only when the related GPIO_FUNC is set to 00)	0x00	0x00	0b000xxxxx It is determined by bits 0 to 4 of the GPIO_Power_Hold register (0x07): 1 = Hold, 0 = State reset
GPIO_IN	GPIO	0x12	[7:5] Reserved [4:0] GPIO4~0	R	—	Real-time input value	—	—	—



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Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
GPIO_DRV	GPIO	0x13	[7:6] Reserved [5] LED EN [4:0] GPIO4~0	R/W	0x1F	1 = Open-drain, 0 = Push-pull	0x1F	0x1F	0b00xxxxxx Among them, LED EN is not affected. GPIO0–GPIO4 are determined by bits 0 to 4 of GPIO_Power_Hold register (0x07): 1 = Hold, 0 = State reset
GPIO_PU/PD_0	GPIO	0x14	[7:6] GPIO3 [5:4] GPIO2 [3:2] GPIO1 [1:0] GPIO0	R/W	0x00	Configured per 2 bits PULL_NO: 00 PULL_UP: 01 PULL_DOWN: 10	0x00	0x00	0bxxxxxxx It is determined by bits 0 to 3 of the GPIO_Power_Hold register (0x07): 1 = Hold, 0 = State reset.
GPIO_PU/PD_1	GPIO	0x15	[7:2] Reserved [1:0] GPIO4	R/W	0x00	Same as above; other bits are reserved.	0x00	0x00	0b000000xx Determined by bit 4 of the GPIO_Power_Hold register (0x07): 1 = Hold, 0 = State reset.
GPIO_FUNC0	GPIO	0x16	[7:6] GPIO3 [5:4] GPIO2 [3:2] GPIO1 [1:0] GPIO0	R/W	0x00	Configured per 2 bits GPIO: 00 IRQ: 01 Special function: 11 Reserved: 10	0x00	0x00	0bxxxxxxx It is determined by bits 0 to 3 of the GPIO_Power_Hold register (0x07): 1 = Hold, 0 = State reset
GPIO_FUNC1	GPIO	0x17	[7:2] Reserved [1:0] GPIO4	R/W	0x00	Same as above; other bits are reserved.	0x00	0x00	0b000000xx Determined by bit 4 of the GPIO_Power_Hold register (0x07): 1 = Hold, 0 = State reset.
GPIO_WAKE_EN	GPIO	0x18	[7:5] Reserved [4:0] GPIO4~0	R/W	0x00	Description: 1 = Enable the corresponding GPIO Wake function; 0 = Disable the corresponding GPIO Wake function (GPIO1 interrupt line conflicts with SDA and cannot be used for the WAKE function. GPIO0 and GPIO2 share one interrupt line and are mutually exclusive. GPIO3 and GPIO4 share one interrupt line and are mutually exclusive).			



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Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
GPIO_WAKE_CFG	GPIO	0x19	[7:5] Reserved [4:0] GPIO4~0	R/W	0x00	GPIO: wake-up 1 = Rising-edge 0 = Falling-edge	—	—	—
—	—	0x1A-0x1F				Reserved	—	—	—
VREF_L	ADC	0x20	[7:0]	R	—	MCU ADC (mV) VREF low 8 bits	—	—	—
VREF_H	ADC	0x21	[7:0]	R	—	MCU ADC (mV) VREF high 8 bits	—	—	—
VBAT_L	ADC	0x22	[7:0]	R	—	BAT Voltage (mV) low 8 bit	—	—	—
VBAT_H	ADC	0x23	[7:0]	R	—	BAT Voltage (mV) high 8 bit	—	—	—
VIN_L	ADC	0x24	[7:0]	R	—	VIN Voltage (mV) low 8 bit	—	—	—
VIN_H	ADC	0x25	[7:0]	R	—	VIN Voltage (mV) high 8 bit	—	—	—
5VOUT_L	ADC	0x26	[7:0]	R	—	5VOUT Voltage (mV) low 8 bit	—	—	—
5VOUT_H	ADC	0x27	[7:0]	R	—	5VOUT Voltage (mV) high 8 bit	—	—	—
ADC_RES_L	ADC	0x28	[7:0]	R	—	ADC result low 8 bit	—	—	—
ADC_RES_H	ADC	0x29	[7:4] Reserved [3:0] Data[11:8]	R	—	ADC result high 4 bit	—	—	—
ADC_CTRL	ADC	0x2A	[7:4] Reserved [3:1] CH_SEL [0] START	R/W	0x00	Description: START = 1 starts the conversion (automatically cleared to 0 after conversion is complete). CH_SEL selects the channel (valid channels are 1, 2, and 6; 1 and 2 correspond to GPIO1 and GPIO2 and take effect only when the corresponding GPIO_FUNC is set to 11; 6 is the internal chip temperature measurement, with the unit in °C).			



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Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
—	—	0x2B-0x2F				Reserved	—	—	—
PWM0_L	PWM	0x30	[7:0] Duty[7:0]	R/W	0x00	PWM0 duty cycle 低 8 bit	—	—	—
PWM0_HC	PWM	0x31	[7:6] Reserved [5] POL [4] EN [3:0] Duty[11:8]	R/W	0x00	EN = 1, Enable POL = 1, Active low PWM1 duty cycle high 8 bits	—	—	—
PWM1_L	PWM	0x32	[7:0] Duty[7:0]	R/W	0x00	PWM1 duty cycle 低 8 bit	—	—	—
PWM1_HC	PWM	0x33	[7:6] Reserved [5] POL [4] EN [3:0] Duty[11:8]	R/W	0x00	EN = 1, Enable POL = 1, Active low PWM1 duty cycle high 8 bits	—	—	—
PWM_FREQ_L	PWM	0x34	[7:0]	R/W	0xF4	PWM frequency low 8 bit	—	—	—
PWM_FREQ_H	PWM	0x35	[7:0]	R/W	0x01	PWM frequency high 8 bit	—	—	—
—	—	0x36-0x37				Reserved	—	—	—
TIM_CNT_BYTE_0	Timer	0x38	[7:0]	R/W	0x00	Timed wake-up counter Byte0 (s)	—	—	—
TIM_CNT_BYTE_1	Timer	0x39	[7:0]	R/W	0x00	Timed wake-up counter Byte1 (s)	—	—	—
TIM_CNT_BYTE_2	Timer	0x3A	[7:0]	R/W	0x00	Timed wake-up counter Byte2 (s)	—	—	—
TIM_CNT_BYTE_3	Timer	0x3B	[7] Reserved [6:0]	R/W	0x00	Timed wake-up counter Byte3 (s)	—	—	—



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Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
TIM_CFG	Timer	0x3C	[7:4] Reserved [3] ARM [2:0] ACTION	R/W	0x00	ARM = 1, starts counting (if TIM_CNT is 0, ARM will be automatically cleared); ACTION: see Table 4.	Cleared to 0 when the timer is triggered	Cleared to 0 when the timer is triggered	0x00
TIM_KEY	Timer	0x3D	[7:0]	W	—	Write 0xA5 to clear and reload	—	—	—
—	—	0x3E-0x3F				Reserved	—	—	—
IRQ Status 1	IRQ	0x40	[7:5] Reserved [4:0] GPIO4~0	R/W	0x00	Description: When the bit of the corresponding GPIO equals 1, it indicates that the level of the corresponding GPIO has changed. The corresponding GPIO level can be read via register 0x06 (except for GPIOs configured as IRQ; other IOs must have the corresponding GPIO_FUNC set to 00 and GPIO_MODE set to 0). In this case, GPIOs set as IRQ will be pulled low and will only be released and pulled high again after the IRQ Status is cleared.			
IRQ Status 2	IRQ	0x41	[7:6] Reserved [5] 电池移除 [4] 电池插入 [3] 5VINOUT 移除 [2] 5VINOUT 插入 [1] 5V IN 移除 [0] 5V IN 插入	R/W	0x00	Description: When the corresponding bit equals 1, it indicates that the corresponding event has occurred. At this time, the GPIO configured as IRQ will be pulled low and will only be released and pulled high again after the IRQ Status is cleared. Note: 1. Battery insertion/removal is only valid when battery charging is disabled; it is invalid when battery charging is enabled. 2. 5VIN/OUT insertion/removal is only valid when 5VIN/OUT is set to INPUT; it is invalid when set to OUTPUT.			
IRQ Status 3	IRQ	0x42	[7:3] Reserved [2] DOUBLE_CLICK [1] WAKEUP [0] SINGLE_CLICK	R/W	0x00	Description: 1. Bit0 is also a reset detection bit. After the PWR_BTN reset function is disabled, a single click of the PWR_BTN will trigger the button single-click interrupt. 2. WAKE_SRC (0x2F) is related to bit1 of IRQ Status 3. That is, if WAKE_SRC (0x2F) is not cleared, IRQ Status 3 bit1 will remain 1. 3. Bit2 is also a power-off detection bit. After the PWR_BTN double-click power-off function is disabled, a double click of the PWR_BTN will trigger the button double-click interrupt.			
IRQ Status 1_Mask	IRQ	0x43	[7:5] Reserved [4:0] GPIO4~0	R/W	0x00	Set corresponding bit to 1 indicates that the interrupt is masked	—	—	—



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Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
IRQ Status 2_Mask	IRQ	0x44	[7:6] Reserved [5] Battery remove [4] Battery add [3] 5VINOUT remove [2] 5VINOUT add [1] 5V IN remove [0] 5V IN add	R/W	0x00	Set corresponding bit to 1 indicates that the interrupt is masked	—	—	—
IRQ Status 3_Mask	IRQ	0x45	[7:3] Reserved [2] Double click [1] Wakeup [0] Click	R/W	0x00	Set corresponding bit to 1 indicates that the interrupt is masked	—	—	—
—	—	0x46-0x47				Reserved	—	—	—
BTN_Status	BTN	0x48	[7] BTN_EVENT [6:1] Reserved [0] BTN_Status	R	0x00	Button status	—	—	—
BTN_CFG_1	BTN	0x49	[7] DL_LOCK [6:5] DBL [4:3] LONG [2:1] SINGLE [0] SINGLE_RESET_DIS	R/W	0x2A	Button Configuration Register 1	—	—	—
BTN_CFG_2	BTN	0x4A	[7:1] Reserved [0] DOUBLE_POWEROFF_DIS	R/W	0x00	Button Configuration Register 2	—	—	—
—	—	0x4B-0x4F				Reserved	—	—	—
NEO_CFG	RGB	0x50	[7] Reserved [6] REFRESH [5:0] LED_CNT	R/W	0x00	Description: NeoPixel count, refresh control for 32 LEDs takes approximately 7 ms. During this time, interrupts are disabled, meaning I ² C communication will be blocked			
—	—	0x51-0x52				Reserved	—	—	—



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Register Name	Type	Addr	Bit	R/W	Default	Description	Reset	Download	Power Off
AW8737A	PULSE	0x53	[7] REFRESH [6:5] NUM [5:0] GPIO	R/W	0x00	GPIO effective value 0 ~ 4	—	—	—
—	—	0x54-0x5F				Reserved	—	—	—
NEO_PLXn_L/H	RGB	0x60-0x9F	[7:0]	R/W	0x00	Up to 32 × RGB565 pixel data, 64 bytes.	—	—	—
RTC_MEM	RTC	0xA0-0xBF	[7:0]	R/W	0x00	32 Byte RTC backup RAM	—	—	—
—	—	0xC0-0xFF				Reserved	—	—	—

Note:

1. RES: Reserved bits
2. Reset includes button reset, command reset, I2C watchdog reset, and timer reset
3. Power-off includes button power-off, command power-off, and timer power-off



IV. Key Register Description

⚠ Register Access Limitation: I2C continuous read/write operations are only supported for specific address ranges (0x00–0x0C, 0x10–0x19, 0x20–0x2A, 0x30–0x35, 0x38–0x3D, 0x40–0x45, 0x48–0x4A, 0x50, 0x53, 0x60–0x9F, 0xA0–0xBF).

Cross-range operations must be performed in multiple transactions.

1 System Registers

(1) Device_ID (0x00) :

- Access: R
- Default: 0x50
- Function: Device Type

(2) Device_Model (0x01):

- Access: R
- Default: 0x20
- Function: Device Model

(3) HW_REV (0x02):

- Access: R
- Default: 0x05
- Function: Hardware Version

(4) SW_REV (0x03):

- Access: R
- Default: 0x06
- Function: Firmware Version

(5) PWR_SRC (0x04):

- Access: R
- Default: None
- Function: Power Source Status
- Bit definition:

[7~3]: Reserved

[2] BAT: Battery valid

[1] 5VINOUT: 5VINOUT valid (only when 5V boost is disabled; when 5V boost is enabled, this bit is 0)

[0] 5VIN: 5VIN valid

Note :

1. Multiple power sources may exist simultaneously. The system determines the current power source status by detecting whether there is voltage on the corresponding ADC pins.



2. When battery charging is enabled but no battery is connected, the voltage measured by the ADC may float and become unstable, causing instability in battery power source detection. It is recommended to disable battery charging when no battery is present.

(6) WAKE_SRC (0x05):

- Access: R/W
- Default: None
- Function: Wake-up Source Flag
- Bit definition:

[7]: Reserved

[6] **5V INOUT**: 5V INOUT insertion wake-up (only when 5V boost is disabled)

[5] **EXT_WAKE**: GPIO WAKE wake-up

[4] **CMD_RST**: Reset command wake-up

[3] **RSTBTN**: Reset button wake-up

[2] **PWRBTN**: Power button wake-up

[1] **VIN**: 5VIN insertion wake-up

[0] **TIM**: Timer wake-up

Note: Writing to this register can only clear the flag bit to ensure valid detection of the wake-up source the next time; it cannot specify a wake-up source.

(7) PWR_CFG (0x06):

- Access: R/W
- Default: 0x17 (0b0001 0111)
- Function: Power switch control
- Bit definition:

[7:5]: Reserved

[4] **LED CONTROL**: 1 = LED EN outputs high level, 0 = LED EN outputs low level

[3] **5VIN/OUT**: 1 = 5V boost output enabled, 0 = 5V boost disabled (external input power can be connected)

[2] **3.3V_LDO_EN**: 1 = Enable 3.3V LDO, 0 = Disable 3.3V LDO

[1] **3.3V_DCDC_EN**: 1 = Enable 3.3V DC/DC, 0 = Disable 3.3V DC/DC

[0] **CHG_EN**: 1 = Enable charging, 0 = Disable charging

(8) HOLD_CFG(0x07):

- Access: R/W
- Default: 0x00
- Function: Hold register, including power hold and GPIO hold



- Bit definition:

[7]: Reserved

[6] 5vin/out: 5vin/out = 1 , 5vin/out power is retained after power-off; 5vin/out = 0, the 5vin/out power is not retained after power-off

[5] ldo_3v3: ldo_3v3 = 1, ldo_3v3 power is retained after power-off; ldo_3v3 = 0, the ldo_3v3 power is not retained after power-off

[4] gpio4: gpio4 = 1 , GPIO4 state is retained after power-off; gpio4 = 0, the state of GPIO4 is reset after power-off

[3] gpio3: gpio3 = 1, GPIO3 state is retained after power-off; gpio3 = 0, the state of GPIO3 is reset after power-off

[2] gpio2: gpio2 = 1, GPIO2 state is retained after power-off; gpio2 = 0, the state of GPIO2 is reset after power-off

[1] gpio1: gpio1 = 1, GPIO1 state is retained after power-off; gpio1 = 0, the state of GPIO1 is reset after power-off

[0] gpio0: gpio0 = 1, GPIO0 state is retained after power-off; gpio0 = 0, the state of GPIO0 is reset after power-off

(9) BATT_LVP (0x08):

- Access: R/W
- Default: 0x40
- Function: Low-voltage threshold register.

When the voltage is lower than the low-voltage threshold, the system will forcibly power off

Low-voltage value calculation: $2.0\text{ V} + \text{reg_value} \times 7.81\text{ mV}$

Note : Power-on recovery conditions: the system can power on again when any one of the following conditions

1. Battery voltage is 100 mV higher than the configured voltage
2. 5VIN is inserted
3. 5VINOUT is inserted (only when the 5V boost is disabled)

(10) I2C_CFG (0x09):

- Access: R/W
- Default: 0x00
- Function: I²C speed configuration and idle sleep configuration
- Bit definition:

[7:5] DBL: Reserved



[4] SPD: 0=100 k, 1=400 k

[3-0] SLP_TO: Specifies how many seconds without I²C communication before the M5PM1 enters sleep; set to 0 to disable this function.

Note :

1. Once the idle sleep function is configured, M5PM1 will not automatically clear this setting and must be cleared manually by the user.
2. After the idle sleep function is configured and successfully triggered, M5PM1 enters sleep mode. If it is woken up via I²C communication, the first I²C transaction is used only for wake-up and will fail. Subsequent communications will operate normally. Note that within 300 ms after the first wake-up communication, if M5PM1 does not receive a complete address, it will enter sleep mode again.

(11) WDT_CNT (0x0A):

- Access: R/W
- Default: 0x00
- Function: Software watchdog, timeout to reset

Note : The watchdog countdown unit is seconds; set to 0 to disable this function.

(12) WDT_KEY (0x0B):

- Access: W
- Default: None
- Function: Feed software watchdog

Note : Write 0xA5 to clear and reload

(13) SYS_CMD (0x0C):

- Access: W
- Default: None
- Function: System Command Register
- Bit definition:

[7:4] KEY: 0xA

[3-2]: Reserved

[1-0] CMD: Command, 01=Power off, 10= Restart, 11=Download

- Write command: KEY=0xA + CMD (01 = Power off, 10 = Restart, 11 = Download)

2 GPIO Register

(1) GPIO_MODE (0x10):

- Access: R/W
- Default: 0x00



- Function: GPIO mode
- Bit definition:
 - [7:5]: Reserved
 - [4:0]: Corresponding GPIO4–GPIO0 direction (1 = Output, 0 = Input)
- Effective condition: GPIO_FUNCx set to 00

(2) GPIO_OUT (0x11):

- Access: R/W
- Default: 0x00
- Function: GPIO output level
- Bit definition:
 - [7:5]: Reserved
 - [4:0]: Corresponding GPIO4–GPIO0 output level (1=High, 0=Low)
- Effective condition:
 - GPIO_FUNCx set to 00; GPIO_MODE set to 1

(3) GPIO_IN (0x12):

- Access: R
- Default: None
- Function: GPIO input status
- Bit definition:
 - [7:5]: Reserved
 - [4:0]: Corresponding GPIO4–GPIO0 input level (1=High, 0=Low)

(4) GPIO_DRV (0x13):

- Access: R/W
- Default: 0x1F
- Function: GPIO output type
- Bit definition:
 - [7:5]: Reserved
 - [4:0]: Corresponding GPIO4–GPIO0 output type (1=Open-drain, 0=Push-pull)

Priority description:

Output type configuration has higher priority than the multiplexing function and does not become invalid when the multiplexing function is enabled or disabled.

For example, when a GPIO is multiplexed as PWM, if GPIO_DRV = 1 (open-drain), the actual output remains in open-drain mode.



(5) GPIO_PU/PD_0 (0x14), GPIO_PU/PD_1 (0x15):

- Access: R/W
- Default: 0x00
- Function: GPIO pull-up / pull-down configuration
- Bit definition: (Each 2 bits control one GPIO)

00: No pull-up / pull-down

01: Pull-up

10: Pull-down

(6) GPIO_FUNC0 (0x16), GPIO_FUNC1 (0x17):

- Access: R/W
- Default: 0x00
- Function: GPIO function
- Bit definition: (Each 2 bits control one GPIO)

00: Standard GPIO;

01: IRQ interrupt

11: Multiplexed function (NeoPixel/ADC/PWM); **10:** Reserved

(7) GPIO_WAKE_EN (0x18):

- Access: R/W
- Default: 0x00
- Function: GPIO wake-up enable
- Bit definition:

[7:5]: Reserved

[4:0]: Corresponding GPIO4-GPIO0 wake enable (1 = Enable, 0 = Disable; GPIO1 is not supported)

Note :

1. GPIO WAKE configuration does not become invalid when entering Download mode, Reset, or Power-off.
2. Whether pull-up or pull-down is enabled for WAKE is controlled by GPIO_PU/PD_0 (0x14) and GPIO_PU/PD_1 (0x15).

(8) GPIO_WAKE_CFG (0x19):

- Access: R/W
- Default: 0x00
- Function: Wakeup configuration
- Bit definition:

[7:5]: Reserved



[4:0]: Corresponding GPIO4–GPIO0 edge configuration (1 = Rising edge, 0 = Falling edge; GPIO1 is not supported).

- Effective condition: Takes effect only after WAKE_EN is enabled.

Note : GPIO WAKE configuration does not become invalid when entering Download mode, Reset, or Power-off.

3 ADC Register

(1) VREF_L (0x20)、VREF_H (0x21) :

- Access: R
- Default: None
- Function: Internal reference voltage VREF

Note : MCU ADC internal reference voltage, unit: mV

(2) VBAT_L (0x22)、VBAT_H (0x23) :

- Access: R
- Default: None
- Function: Battery voltage VBAT

Note : Battery voltage, unit: mV

(3) VIN_L (0x24)、VIN_H (0x25) :

- Access: R
- Default: None
- Function: 5V input voltage VIN

Note : 5VIN voltage, unit: mV

(4) 5VOUT_L (0x26)、5VOUT_H (0x27) :

- Access: R
- Default: None
- Function: 5V output voltage 5VOUT

Note : 5V output voltage, unit: mV

(5) ADC_RES_L (0x28)、ADC_RES_H (0x29):

- Access: R
- Default: None
- Function: Channel conversion value
- Note :

1. Combined into 12-bit data ([11:0]). When the channel selection is 1 or 2, it represents a 12-bit ADC result with a range of 0–0xFFF.



2. When the channel selection is 6, it represents the internal chip temperature, in °C.

(6) ADC_CTRL (0x2A):

- Access: R/W
- Default: 0x00
- Function: ADC Conversion Register
- Bit definition:

[7:4]: Reserved

[3:1] CH_SEL: Channel selection (1 = GPIO1, 2 = GPIO2, 6 = Internal temperature channel)

[0] START: Write 1 to start conversion (automatically cleared after completion)

4 PWM Control Register

(1) PWM0_L (0x30) :

- Access: R/W
- Default: 0x00
- Function: PWM Register (Duty Cycle Low 8 Bits)

(2) PWM0_HC (0x31) :

- Access: R/W
- Default: 0x00
- Function: PWM Register
- Bit definition:

[7:6]: Reserved

[5] POL: Polarity (1 = Active low)

[4] EN: Enable (1 = Start)

[3:0] Duty[11:8]: Duty cycle high 4 bits (combined with PWM0_L to form a 12-bit duty cycle)

(3) PWM1_L (0x32) :

- Access: R/W
- Default: 0x00
- Function: PWM Register (Duty Cycle Low 8 Bits)

(4) PWM1_HC (0x33) :

- Access: R/W
- Default: 0x00
- Function: PWM Register
- Bit definition:



[7:6]: Reserved

[5] **POL**: Polarity (1 = Active low)

[4] **EN**: Enable (1 = Start)

[3:0] **Duty**[11:8]: Duty cycle high 4 bits (combined with PWM1_L to form a 12-bit duty cycle)

(5) **PWM_FREQ_L** (0x34)、**PWM_FREQ_H** (0x35):

- Access: R/W
- Default: 0xF4、0x01
- Function: Configure PWM frequency, unit: Hz

5 System Timer

(1) **TIM_CNT_BYTE_0~3** (0x38-0x3B):

- Access: R/W
- Default: 0x00
- Function: Timed wake-up counter, unit: seconds, range: 0–214748364

(2) **TIM_CFG** (0x3C):

- Access: R/W
- Default: 0x00
- Function: Timer function configuration
- Bit definition:

[7:6]: Reserved

[3] **ARM**: 1=Start the timer (automatically cleared when the count reaches 0)

[2:0] **ACTION**: Timer action (see Table 4)

Note :

1. System power-off and re-power-on will clear the TIM_CFG register.
2. After the wake-up flag is set, system power-on, system restart, or system power-off takes effect once, the TIM_CFG register will be cleared.

Table 4 Timer Action Truth Table

ACTION	Function
0	Stop counter
1	Set wake-up flag
10	System restart
11	System power on
100	System power off

(3) **TIM_KEY** (0x3D):

- Access: W
- Default: None
- Function: Reload timer (write 0xA5)



6 Interrupt and Wake-up Control

(1) IRQ Status 1 (0x40):

- Access: R/W
- Default: 0x00
- Function: IRQ Register 1
- Bit definition:

[7:5]: Reserved

[4:0]: Indicates the level change status of GPIO4–GPIO0; a value of 1 indicates that the corresponding GPIO has a level change.

Note : This register can only be cleared by the user; setting bits to 1 is performed automatically by the system.

(2) IRQ Status 2 (0x41):

- Access: R/W
- Default: 0x00
- Function: IRQ Register 2
- Bit definition: Corresponding bit value of 1 indicates that the event has occurred

[7:6]: Reserved

[5] **Battery Remove**: Battery removed, BAT voltage $\geq$ 2400 mV $\rightarrow$ BAT voltage < 2400 mV

[4] **Battery Add**: Battery inserted, BAT voltage $\leq$ 2400 mV $\rightarrow$ BAT voltage > 2400 mV

[3] **5VINOUT Remove**: 5VIN/OUT inserted, 5VINOUT voltage $\leq$ 2400 mV $\rightarrow$ 5VINOUT voltage > 2400 mV

[2] **5VINOUT Add**: 5VIN/OUT inserted, 5VINOUT voltage $\leq$ 2400 mV $\rightarrow$ 5VINOUT voltage > 2400 mV

[1] **5VIN Remove**: 5VIN removed, 5VIN voltage $\geq$ 2400 mV $\rightarrow$ 5VIN voltage < 2400 mV

[0] **5VIN Add**: 5VIN inserted, 5VIN voltage $\leq$ 2400 mV $\rightarrow$ 5VIN voltage > 2400 mV

Note : This register can only be cleared by the user; setting bits to 1 is performed automatically by the system.

- Effective condition:

1. Battery insertion/removal events are only valid when CHG_EN = 0 (charging disabled).
2. 5VINOUT insertion/removal events are only valid when 5VIN/OUT = 0 (INPUT mode).

(3) IRQ Status 3 (0x42):

- Access: R/W
- Default: 0x00
- Function: IRQ Register 3
- Bit definition: Corresponding bit to 1 indicates that the event has occurred

[7:3]: Reserved



[2] DOUBLE_CLICK: Button double-click

[1] WAKEUP: Power-on

[0] SINGLE_CLICK: Button single-click

Note : This register can only be cleared by the user; setting bits to 1 is performed automatically by the system.

- Effective condition:

1. Bit0 is also a reset detection bit. After the PWR_BTN reset function is disabled, a single click of the PWR_BTN will trigger the button single-click interrupt.

2. WAKE_SRC (0x2F) is related to bit1 of IRQ Status 3. That is, if WAKE_SRC (0x2F) is not cleared, IRQ Status 3 bit1 will remain 1.

3. Bit2 is also a power-off detection bit. After the PWR_BTN double-click power-off function is disabled, a double click of the PWR_BTN will trigger the button double-click interrupt.

(4) IRQ Status 1 Mask(0x43):

- Access: R/W

- Default: 0x00

- Function: IRQ Mask Register 1

- Bit definition:

[7:5]: Reserved

[4:0]: Corresponding GPIO4-GPIO0

Note : Setting the corresponding bit to 1 masks the interrupt

(5) IRQ Status 2 Mask(0x44):

- Access: R/W

- Default: 0x00

- Function: IRQ Mask Register 2

- Bit definition:

[7:6]: Reserved

[5] Battery Remove: Battery removed, BAT voltage $\geq$ 2400 mV $\rightarrow$ BAT voltage < 2400 mV

[4] Battery Add: Battery inserted, BAT voltage $\leq$ 2400 mV $\rightarrow$ BAT voltage > 2400 mV

[3] 5VINOUT Remove: 5VIN/OUT inserted, 5VINOUT voltage $\leq$ 2400 mV $\rightarrow$ 5VINOUT voltage > 2400 mV

[2] 5VINOUT Add: 5VIN/OUT inserted, 5VINOUT voltage $\leq$ 2400 mV $\rightarrow$ 5VINOUT voltage > 2400 mV

[1] 5VIN Remove: 5VIN removed, 5VIN voltage $\geq$ 2400 mV $\rightarrow$ 5VIN voltage < 2400 mV

[0] 5VIN Add: 5VIN inserted, 5VIN voltage $\leq$ 2400 mV $\rightarrow$ 5VIN voltage > 2400 mV

Note : Setting the corresponding bit to 1 masks the interrupt

**(6) IRQ Status 3 Mask (0x45):**

- Access: R/W
- Default: 0x00
- Function: IRQ Mask Register 3
- Bit definition:

[7:3]: Reserved

[2] DOUBLE_CLICK: Button double-click

[1] WAKEUP: Power-on

[0] SINGLE_CLICK: Button single-click

Note : Setting the corresponding bit to 1 masks the interrupt

IRQ Attention:

1. When no GPIO is configured as an IRQ pin, IRQ Status 1 (0x40), IRQ Status 2 (0x41), and IRQ Status 3 (0x42) will be cleared.
2. During Reset and when entering Download mode, GPIOs are reset. At this time, no GPIO is configured as an IRQ pin, so IRQ Status 1 (0x40), IRQ Status 2 (0x41), and IRQ Status 3 (0x42) will be cleared. If you need to use the IRQ Status 3 (0x42) Wakeup IRQ in this situation, do not clear the WAKE_SRC (0x05) register until the GPIO is configured as an IRQ pin again; otherwise, the IRQ Status 3 (0x42) Wakeup IRQ cannot be triggered.

7 Button Configuration**(1) BTN_Status (0x48) :**

- Access: R
- Default: None
- Function: Button status
- Bit definition:

[7] BTN_Event: Press event, 1 = Button has been pressed, 0 = Button has not been pressed; automatically cleared after reading, set to 1 automatically by the system

[6:1]: Reserved

[0] BTN_Status: Press status, 1 = Pressed, 0 = Released

(2) BTN_CFG (0x49):

- Access: R/W
- Default: 0x2A
- Function: Button Configuration 1
- Bit definition:



[7] DL_LOCK: DL_LOCK = 1, Disable Download mode

[6:5] DBL: Double-click, 00 = 125 ms, 01 = 250 ms, 10 = 500 ms, 11 = 1 s

[4:3] LONG: Long press, 00 = 1 s, 01=2 s, 10=3 s, 11=4 s

[2:1] SINGLE: Single-click, 00 = 125 ms, 01 = 250 ms, 10 = 500 ms, 11 = 1 s

[0] SINGLE_RESET_DIS: SINGLE_RESET_DIS = 1 — Disable single-click reset; SINGLE_RESET_DIS = 0 — Enable single-click reset

(3) BTN_CFG_2 (0x4A):

- Access: R/W
- Default: 0x00
- Function: Button Configuration 2
- Bit definition:

[7:1]: Reserved

[0] DOUBLE_POWEROFF_DIS:

DOUBLE_POWEROFF_DIS = 1 — Disable double-click power-off

DOUBLE_POWEROFF_DIS = 0 — Enable double-click power-off

8 NeoPixel Control Module

NEO_CFG (0x50):

- Access: R/W
- Default: 0x00
- Function: RGB Configuration
- Bit definition:

[7]: Reserved

[6] REFRESH: Refresh control (refreshing 32 LEDs takes approximately 7 ms, during which I²C interrupts are disabled)

[5:0] LED_CNT: LED count (0–32, 0 = Disable driver)

9 AW8737A PULSE

PULSE_CTRL (0x53):

- Access: R/W
- Default: 0x00
- Function: AW8737A pulse output control
- Bit definition:

[7] REFRESH: 1 = Refresh; 0 = No refresh

[6:5] NUM: Value range 0–3, actual output of 0–3 pulse(s)



[4:0] GPIO: Value range 0–4, corresponding to hardware GPIO0 ~ GPIO4

Note :

1. After successful configuration, the corresponding GPIO is set to output mode and the GPIO output is modulated with the corresponding pulse.
2. If open-drain output mode is used, an external pull-up is required; otherwise, configure the corresponding GPIO to push-pull output mode first.

10 NEO Buffer

NEO_PIXn_L/H (0x60-0x9F):

- Access: R/W
- Default: 0x00
- Function: RGB buffer

Note : Each LED occupies 2 bytes (RGB565 format), stored sequentially (PIX0_L = 0x36, PIX0_H = 0x37, ..., PIX31_H = 0x75).

11 RTC Buffer

RTC_MEM[0:31] (0xA0-0xBF):

- Access: R/W
- Default: 0x00
- Function: RTC buffer

Note : 32-byte power-retention RAM (power retention here refers to ESP32 power-off retention; M5PM1 power-off does not retain data).



V. Additional Function Description

1 ADC

(1) Flow:

Write ADC_CTRL to select the channel and start (START = 1) → wait until BUSY = 0 → read ADC_D_H/L.

(2) Usage of Internal ADC Channels

● 5VOUT:

Low-voltage threshold detection

5VOUT power insertion wake-up

5VOUT power removal and insertion detection

● **BAT**: Low-voltage threshold detection, battery removal and insertion detection

● **5VOUT**: Low-voltage threshold detection, 5VIN removal and insertion detection

Attention:

1. 5VOUT has both input and output functions. Before enabling the 5VOUT boost, the 5VOUT voltage must be detected first to ensure that there is no external power input; otherwise, it should remain in input mode.

2. All three channels have specific functions and must be connected strictly according to their intended usage.

2 PWM Output

Duty cycle = $(\text{DUTY}[11:0] / 0\text{xFFF}) \times 100\%$. The frequency is set by the PWM_FREQ register. The high byte and low byte of the value are located in two registers and must be written at the same time; otherwise, two different duty cycles or frequencies may be set within a short period.

3 PWR_BTN Button

Single click for reset, double click for power-off, long press to enter Download mode. The specific timing intervals are determined by the BTN_CFG_1 register.

(1) Power-off

After power-off, the 3.3V DC/DC is turned off and LED_EN is turned off, but the charging enable is not affected.

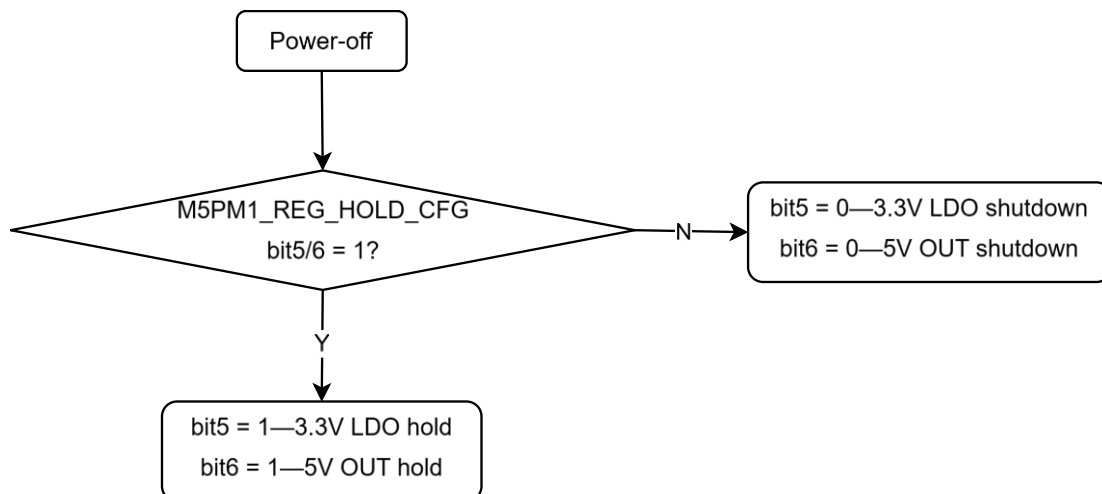


Figure 2 Power-off Flowchart



(2) Download Mode

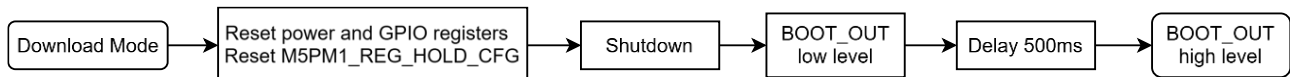


Figure 3 Download Flowchart

(3) Reset

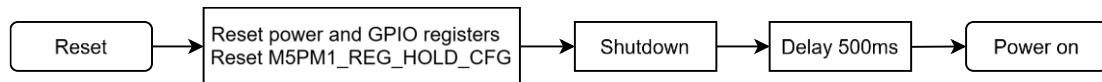


Figure 4 Reset Flowchart

4 LED Indicator

(1) Flow:

Set the LED count (LED_CFG[5:0]) → Write LED_RAM (RGB565 format) → Trigger REFRESH = 1.

(2) LED indication status under different conditions

- **Reset:** LED flashes once
- **Download mode:** LED flashes once every 500 ms
- **Button reset disabled and GPIO IRQ Function enabled:** LED flashes once every 200 ms
- **Button double-click power-off disabled and GPIO IRQ Function enabled:** LED flashes once every 100 ms

5 Low Voltage Protection

The low-voltage threshold is determined by the BATT_LVP (0x08) register. When neither 5VIN nor 5VINOUT is inserted and the battery voltage falls below the threshold, the system will automatically power off.

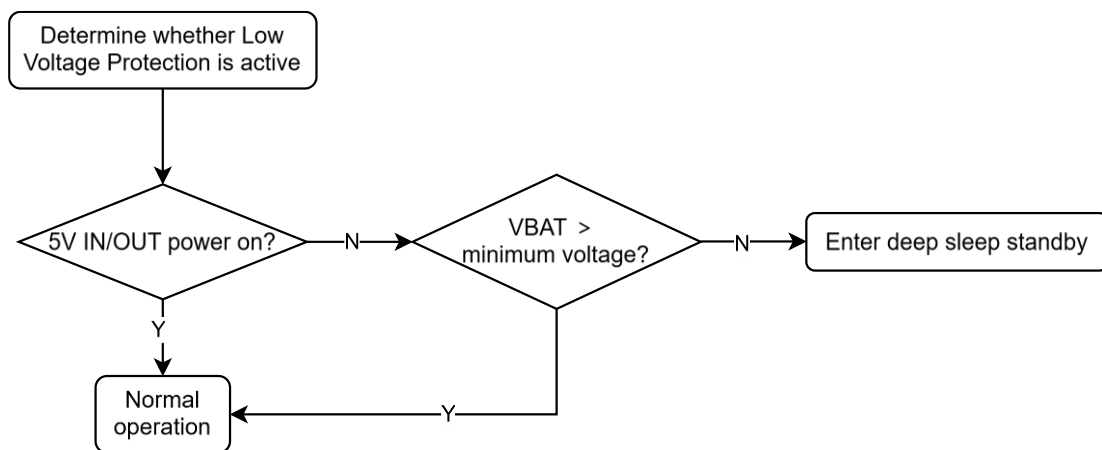


Figure 5 Low-Voltage Detection Flowchart

Note : When VIN is not inserted, the BAT voltage must be greater than the BATT_LVP voltage + 100 mV to exit the low-voltage standby loop.

6 I²C Idle Sleep

Configure SLEEP[3:0] to set the idle sleep time; 0 = Disable sleep.

The complete sleep procedure is as follows:

- Enable PWR_BTN external interrupt
- Enable 5VIN external interrupt



- Enable 5VINOUT external interrupt
- Enable SDA external interrupt
- Configure the timer to wake up every 100 ms (this interval cannot be changed)
- Reset the ADC peripheral
- Set I²C idle sleep flag to 1

Note : When the PWM function is enabled, I²C idle sleep is invalid; when entering Download mode, I²C idle sleep is invalid.

7 Interrupt Wake-up

(1) SDA Wake-up

In sleep mode, if there is I²C communication activity on SDA, M5PM1 will first disable the SDA and PWR_BTN external interrupts, then reinitialize the I²C communication configuration. After that, there will be a 300 ms non-blocking delay. During this period, if the PWR_BTN is pressed, it is considered a PWR_BTN wake-up; otherwise, the system waits for the next I²C communication signal and checks whether the I²C address matches. If it matches, wake-up is successful; otherwise, the system continues sleeping.

(2) PWR_BTN Wake-up

In sleep mode, if the PWR_BTN is pressed, wake-up is successful. The SDA and PWR_BTN external interrupts will be disabled, and the I²C communication configuration will be reinitialized.

(3) 5VIN Wake-up (same as (2))

(4) 5VINOUT Wake-up (same as (2))

Note : The above wake-up mechanisms are based on a timer waking up every 100 ms. This is a firmware-fixed process and cannot be changed.

8 IRQ Handling

(1) When any GPIO is configured as an IRQ pin, the GPIO configured as IRQ will be pulled low when IRQ Status 1 (0x40), IRQ Status 2 (0x41), or IRQ Status 3 (0x42) is non-zero. It will be released and pulled high only when IRQ Status 1 (0x40), IRQ Status 2 (0x41), and IRQ Status 3 (0x42) are all cleared.

(2) When any GPIO is configured as an IRQ pin, the non-IRQ IOs (GPIO_FUNC set to 00 and GPIO_MODE set to 0) will be scanned. When the level of a scanned IO changes, the corresponding bit in IRQ Status 1 (0x40) will be set to 1.

(3) When any of the five GPIOs is configured as an IRQ pin, power events will be scanned. When a power event occurs, the corresponding bit in IRQ Status 2 (0x41) will be set to 1.

Power events are as follows:

- Battery Remove: Battery removed, BAT voltage $\geq$ 2400 mV $\rightarrow$ BAT voltage < 2400 mV
- Battery Add: Battery inserted, BAT voltage $\leq$ 2400 mV $\rightarrow$ BAT voltage > 2400 mV
- 5VINOUT Remove: 5VIN/OUT removed, 5VINOUT voltage $\geq$ 2400 mV $\rightarrow$ 5VINOUT voltage < 2400 mV
- 5VINOUT Add: 5VIN/OUT inserted, 5VINOUT voltage $\leq$ 2400 mV $\rightarrow$ 5VINOUT voltage > 2400 mV



- 5VIN Remove: 5VIN removed, 5VIN voltage ≥ 2400 mV $\rightarrow$ 5VIN voltage < 2400 mV
- 5VIN Add: 5VIN inserted, 5VIN voltage ≤ 2400 mV $\rightarrow$ 5VIN voltage > 2400 mV

(4) After Reset or entering Download mode, the GPIOs will be reset. At this time, no GPIO is configured as an IRQ pin, and IRQ Status 1 (0x40), IRQ Status 2 (0x41), and IRQ Status 3 (0x42) will be cleared. **If you need to use the IRQ Status 3 (0x42) Wakeup IRQ, do not clear the WAKE_SRC (0x04) register until the GPIO is configured as an IRQ pin again; otherwise, the IRQ Status 3 (0x42) Wakeup IRQ cannot be triggered.**



VI. Use Examples

1 GPIO Wake-up

(1) Rising-edge wake-up control

```
pm1.gpioSetMode(M5PM1_GPIO_NUM_0, M5PM1_GPIO_MODE_INPUT);    // Set GPIO0 to input mode
pm1.gpioSetPull(M5PM1_GPIO_NUM_0, M5PM1_GPIO_PULL_DOWN);      // Set GPIO0 pull-down
(gpioSetPull can be ignored if external pull-down is present)
pm1.gpioSetWakeEnable(M5PM1_GPIO_NUM_0, true);                 // Enable GPIO0 wake-up
pm1.gpioSetWakeEdge(M5PM1_GPIO_NUM_0, M5PM1_GPIO_WAKE_RISING); // Set GPIO0 rising-edge wake-up
```

(2) Falling-edge wake-up control

```
pm1.gpioSetMode(M5PM1_GPIO_NUM_0, M5PM1_GPIO_MODE_INPUT);    // Set GPIO0 to input mode
pm1.gpioSetPull(M5PM1_GPIO_NUM_0, M5PM1_GPIO_PULL_UP);        // Set GPIO0 pull-up
(gpioSetPull can be ignored if external pull-up is present)
pm1.gpioSetWakeEnable(M5PM1_GPIO_NUM_0, true);                 // Enable GPIO0 wake-up
pm1.gpioSetWakeEdge(M5PM1_GPIO_NUM_0, M5PM1_GPIO_WAKE_FALLING); // Set GPIO0 falling-edge wake-up
```

(3) Attention

GPIO0, GPIO2, GPIO3, and GPIO4 support wake-up. However, GPIO0 and GPIO2 are mutually exclusive, and GPIO3 and GPIO4 are mutually exclusive and cannot be used at the same time.

2 RGB

(1) RGB control

```
pm1.gpioSetFunc(M5PM1_GPIO_NUM_0, M5PM1_GPIO_FUNC_OTHER);    // Set GPIO0 to multiplexed function
pm1.gpioSetDrive(M5PM1_GPIO_NUM_0, M5PM1_GPIO_DRIVE_PUSH_PULL); // Set GPIO0 to push-pull output mode

m5pm1_rgb_t rgb_red = {255, 0, 0};
m5pm1_rgb_t rgb_green = {0, 255, 0};
m5pm1_rgb_t rgb_blue = {0, 0, 255};
m5pm1_rgb_t rgb_array[3] = { rgb_red, rgb_green, rgb_blue };

pm1.setLeds(&rgb_red, 1, 3, true);    // Set 3 RGB LEDs to red and refresh
// Appropriate delay
pm1.setLeds(&rgb_green, 1, 3, true);  // Set 3 RGB LEDs to green and refresh
// Appropriate delay
pm1.setLeds(&rgb_blue, 1, 3, true);    // Set 3 RGB LEDs to blue and refresh
// Appropriate delay
pm1.setLeds(rgb_array, 3, 1, true);    // Set 3 RGB LEDs to red, green, blue in sequence and refresh
// Appropriate delay
```

(2) Attention

- 1) When GPIO0 is set to a multiplexed function, the system switches the clock to 24 MHz to prepare for RGB timing refresh. Switching the clock will reset I²C, so you must wait for a period of time before performing I²C communication; otherwise, communication will fail.
- 2) The output mode strictly follows GPIO_DRV (0x13). If there is no external pull-up, GPIO must be configured as push-pull to output correct RGB timing.
- 3) Interrupts are disabled during RGB output. Refreshing 32 LEDs takes about 7 ms, during which I²C communication cannot be performed. After refreshing RGB timing, wait for a period of time before communication.
- 4) Only GPIO0 supports the RGB function.



3 ADC

(1) ADC usage example

```
pm1.gpioSetFunc(M5PM1_GPIO_NUM_1, M5PM1_GPIO_FUNC_OTHER); // Set GPIO1 to multiplexed function
pm1.gpioSetFunc(M5PM1_GPIO_NUM_2, M5PM1_GPIO_FUNC_OTHER); // Set GPIO2 to multiplexed function
uint16_t gpio1_adc_value = 0;
uint16_t gpio2_adc_value = 0;
uint16_t temp_value = 0;
uint16_t vref_value = 0;
pm1.analogRead(M5PM1_ADC_CH_1, &gpio1_adc_value); // Read ADC value of GPIO1
pm1.analogRead(M5PM1_ADC_CH_2, &gpio2_adc_value); // Read ADC value of GPIO2
pm1.readTemperature(M5PM1_ADC_CH_TEMP, &temp_value); // Read internal MCU temperature
pm1.readVref(&vref_value); // Read reference voltage for ADC calibration
uint16_t gpio1_volt = (gpio1_adc_value * vref_value) / 4096; // Calculate actual voltage of GPIO1
uint16_t gpio2_volt = (gpio2_adc_value * vref_value) / 4096; // Calculate actual voltage of GPIO2
```

(2) Attention

- 1) The ADC channel input voltage must not exceed 3.3 V; otherwise, the ADC module may malfunction.
- 2) The actual voltage is calculated using the reference voltage VREF.

4 PWM

(1) PWM usage example

```
pm1.gpioSetFunc(M5PM1_GPIO_NUM_3, M5PM1_GPIO_FUNC_OTHER); // Set GPIO3 to multiplexed function
pm1.gpioSetFunc(M5PM1_GPIO_NUM_4, M5PM1_GPIO_FUNC_OTHER); // Set GPIO4 to multiplexed function
pm1.gpioSetDrive(M5PM1_GPIO_NUM_3, M5PM1_GPIO_DRIVE_PUSH_PULL); // Set GPIO3 to push-pull output mode
pm1.gpioSetDrive(M5PM1_GPIO_NUM_4, M5PM1_GPIO_DRIVE_PUSH_PULL); // Set GPIO4 to push-pull output mode
pm1.setPwmFrequency(20000); // Set PWM frequency to 20 kHz
pm1.setPwmDuty(M5PM1_PWM_CH_0, 50, false, true); // Set 50% duty cycle PWM on GPIO3
pm1.setPwmDuty(M5PM1_PWM_CH_1, 50, false, true); // Set 50% duty cycle PWM on GPIO4
```

(2) Attention

- 1) The two PWM channels are controlled by the same timer, so both channels share the same frequency.
- 2) Duty cycle range: 0–100%.
- 3) If there is no external pull-up, GPIO must be configured as push-pull to output PWM correctly.

5 TIM

(1) System reboot

```
pm1.timerSet(10, M5PM1_TIM_ACTION_REBOOT); // Reboot after 10 seconds
```

(2) System power on

```
pm1.timerSet(10, M5PM1_TIM_ACTION_POWERON); // Power on after 10 seconds;
if the system is powered off, it can automatically power on when the timer expires
```

(3) System power off

```
pm1.timerSet(10, M5PM1_TIM_ACTION_POWEROFF); // Power off after 10 seconds
```



Appendix

Firmware Modification History		
Version	Date	Change Description
HW:2 / SW:1	2025-06-30	Initial version
HW:3 / SW:2	2025-07-23	1. According to the hardware revision, modify the I2C pins and BOOT_OUT pin, adjust the ADC voltage divider ratio, and change CHG_EN to push-pull output with the control logic modified to high = enable and low = disable. 2. Change the default GPIO output type to open-drain. 3. Change the system timer TIM_CNT to 31 bits and adjust the register mapping. 4. Add 5VINOUT wake-up support (only when the 5V boost is disabled).
HW:4 / SW:3	2025-09-01	1. The maximum time limit of the 32-bit timer is 214,748,364 seconds. 2. Added the GPIO_Power_Hold register (0x33). When the corresponding bit is set to 1, the state of the corresponding GPIO or LDO will be retained after power-off. 3. During power-off, GPIOs will be restored to the default state of input with no pull-up/down (if the corresponding bit in the GPIO_Power_Hold register is set to 1, the corresponding GPIO state will be retained after power-off). The power state will be restored to the default state (charging control is not affected; if the corresponding bit in the GPIO_Power_Hold register is set to 1, the corresponding LDO state will be retained after power-off). 4. When entering Download mode, the GPIO_Power_Hold register will be reset to 0, the I2C watchdog will stop, I2C idle sleep will stop, the user timer will stop, GPIOs will be restored to the default input with no pull-up/down state, and the power state will be restored to the default state. 5. When entering Reset mode (including button reset, command reset, I2C watchdog reset, and user timer reset), the GPIO_Power_Hold register will be reset to 0, GPIOs will be restored to the default input with no pull-up/down state, and the power state will be restored to the default state. 6. Added LED indications: a. During reset, the LED flashes once; b. In Download mode, the LED flashes at 500 ms intervals; c. When button reset is disabled and a GPIO has IRQ enabled, the LED flashes at 200 ms intervals; d. When button double-click power-off is disabled and a GPIO has IRQ enabled, the LED flashes at 100 ms intervals. 7. Added IRQ Status 3 (0x23). Bit0 is the reset interrupt. After button reset is disabled, a single click of the PWR_BTN triggers this interrupt. Bit1 is the wake-up interrupt. On power-on or reset, the corresponding bit in WAKE_SRC (0x2F) is set, and bit1 is also set (note that WAKE_SRC (0x2F) is related to bit1 of IRQ Status 3; if WAKE_SRC (0x2F) is not cleared, IRQ Status 3 bit1 will remain 1). Bit2 is the btn_dl_click interrupt. After button double-click power-off is disabled, a double click of the PWR_BTN triggers this interrupt. 8. Added IRQ Mask registers to allow masking of specific interrupts as required. 9. Charging is enabled by default. 10. Added bit0 to BTN_CFG: SINGLE_RESET_DIS. SINGLE_RESET_DIS = 1 disables single-click reset; SINGLE_RESET_DIS = 0 enables single-click reset.



Firmware Modification History		
Version	Date	Change Description
		11. Added register BTN_CFG_2 (0x31): bit0-DOUBLE_POWEROFF_DIS. DOUBLE_POWEROFF_DIS = 1 disables double-click power-off; DOUBLE_POWEROFF_DIS = 0 enables double-click power-off.
HW:5/ SW:4	2025-09-17	1. Added AW8737A pulse-width modulation functionality. 2. Updated the firmware version to V4.
HW:5 / SW:5	2025-11-04	1. Added AW8737A pulse-width modulation output functionality. 2. Added DCDC_5V hold functionality. 3. Modified the watchdog to be disabled by default.
HW:5 / SW:6	2025-12-13	1. Updated the register map. 2. Added the BTN_Status (0x48) register. 3. Removed the UID register and added Device_ID and Device_Model.
HW:5 / SW:S	2026-01-06	1. Modified the timer function so that the timer configuration takes effect once and is automatically cleared; the timer configuration is also cleared after power-off. 2. Adjusted the invalid threshold for USB and 5VOUT to 4V.

Document Revision History		
Version	Version	Version
1.0	2025-06-30	Initial version
1.1	2025-07-23	Hardware revision: 3, Firmware major version: 2
1.2	2025-08-04	Revised some incorrect descriptions
1.3	2025-09-01	Hardware revision: 4, Firmware major version: 3
1.4	2025-09-10	Added detailed explanations for key registers
1.5	2025-09-17	Hardware revision: 5, Firmware major version: 4
1.6	2025-11-04	Hardware revision: 5, Firmware major version: 5
1.7	2025-12-13	Hardware revision: 5, Firmware major version: 6, Added use cases
1.8	2026-01-06	Hardware revision: 5, Firmware major version: S
1.9	2026-01-22	Revised some incorrect descriptions